TCS NQT FORMULA CHEAT SHEET

Comprehensive Quick Reference Guide for Exam Preparation

NUMERICAL APTITUDE

Number System

- Cyclicity lengths: {0,1,5,6} –> 1, {4,9} –> 2, {2,3,7,8} –> 4
- Unit digit of a^b = Unit digit of [(last digit of a) ^ (b mod cyclicity)]
- If b mod cyclicity == 0, use the cyclicity length as the effective exponent.
- HCF(a, b) × LCM(a, b) = a × b
- If d = k × q (k is a factor of d), then N mod k = (N mod d) mod k

Percentages

- Percentage Value = $\frac{\text{Part}}{\text{Base}} \times 100$
- Net Successive Change (a% then b%) = $a + b + (a \times \frac{b}{100})$
- If A is x% more than B, then B is less than A by = $\frac{x}{100 + x} \times 100$
- If A is x% less than B, then B is more than A by = $\frac{x}{100 - x} \times 100$
- $\frac{\text{Population}}{\text{Depreciated Value after n years}} = \text{Initial} \times (\frac{1 \pm r}{100})^n$

Profit, Loss & Discount

- Profit % = $\frac{SP - CP}{CP} \times 100$
- Loss% = $\frac{CP - SP}{CP} \times 100$
- Selling Price (SP) after Discount = MP × $(1 - \frac{\text{Discount\%}}{100})$
- Net Profit % after Mark–up (x%) and Discount (y%) = $x - y - (x \times \frac{y}{100})$
- If CP of m articles = SP of n articles, then Gain% = $\frac{m - n}{n} \times 100$
- Two items sold at identical SP (one at +x%, one at –x%) results in an overall Loss% = $(\frac{x}{10})^2$

Ratio, Proportion & Age Problems

- If A:B = p:q and B:C = r:s, then combined ratio A:B:C = (p×r) : (q×r) : (q×s)
- Product of Extremes = Product of Means ($\frac{\text{If } a}{b} = \frac{c}{d}$, then a × d = b × c)
- Age matching: If ratio changes from a:b to c:d after x years, solve: $\frac{at + x}{bt + x} = \frac{c}{d}$

Speed, Distance & Time

- Speed = $\frac{\text{Distance}}{\text{Time}}$
- Conversion: $\frac{\text{km/h to m}}{\text{s}} = \frac{\text{Multiply by 5}}{18}$ | $\frac{\text{m/s to km}}{\text{h}} = \frac{\text{Multiply by 18}}{5}$
- Train crossing stationary object $\frac{\text{pole}}{\text{man}}$: Distance = Length of Train
- $\frac{\text{Train crossing platform}}{\text{bridge}}$: Distance = Length of Train + Length of Platform
- Relative Speed (Opposite Directions) = Speed_A + Speed_B
- Relative Speed (Same Direction) = Speed_A – Speed_B
- Average Speed (equal split distance covered at speeds a and b) = $\frac{2ab}{a + b}$
- Downstream Speed (d) = Boat Speed (b) + Stream Speed (s)
- Upstream Speed (u) = Boat Speed (b) – Stream Speed (s)
- Boat Speed in still water (b) = $\frac{d + u}{2}$ | Stream Speed (s) = $\frac{d - u}{2}$

Time & Work

- If a person completes work in D days, their 1–day work rate = $\frac{1}{D}$
- If Work is constant: Days taken is inversely proportional to Efficiency (E1 × D1 = E2 × D2)
- Pairwise rates combined: 2 × (A + B + C) = (A + B) + (B + C) + (C + A)
- Net Pipe Filling Rate = Sum of Fill Rates – Sum of Drain Rates
- $\frac{\text{Chain Rule}}{\text{Mandays formula}} : \frac{M1 \times D1 \times H1}{W1} = \frac{M2 \times D2 \times H2}{W2}$

Simple & Compound Interest

- Simple Interest (SI) = $\frac{P \times R \times T}{100}$
- Compound Interest Amount (A) = $P \times (1 + \frac{R}{100})^T$
- Difference between CI and SI for 2 Years = $P \times (\frac{R}{100})^2$
- Difference between CI and SI for 3 Years = $P \times \frac{R}{100})^2 \times (3 + \frac{R}{100}$
- Sum doubles itself under SI in t years at Rate (R) = $\frac{100}{t}$

Averages, Mixtures & Alligation

- Average = $\frac{\text{Total Sum}}{\text{Count of Elements}}$
- Corrected Average = $\frac{\text{Original Sum} - \text{Incorrect Value} + \text{Correct Value}}{\text{Count}}$
- Alligation Ratio $\frac{\text{Quantity of Cheaper}}{\text{Quantity of Dearer}} = \frac{\text{Dearer Price} - \text{Mean Price}}{\text{Mean Price} - \text{Cheaper Price}}$
- Pure Liquid Remaining after n structural replacements = Initial Volume × $(1 - \frac{\text{Quantity Removed}}{\text{Initial Volume}})^n$

Permutation, Combination & Probability

- Permutations (Order matters): $nPr = \frac{n!}{(n-r)!}$
- Combinations (Order does not matter): $nCr = \frac{n!}{r! \times (n-r)!}$
- Arrangement of n items where an item repeats p times = $\frac{n!}{p!}$
- Circular arrangement of n distinct objects = $(n-1)!$
- Probability $P(E) = \frac{\text{Favorable Outcomes}}{\text{Total Sample Space Outcomes}}$

Progressions

- Arithmetic Progression nth term: $a_n = a + (n-1)d$
- Arithmetic Progression Sum: $S_n = \frac{n}{2} \times [\, 2a + (n-1)d \,]$ or $\frac{n}{2} \times (a + l)$
- Geometric Progression nth term: $a_n = a \times r^{(n-1)}$
- Geometric Progression Sum ($r > 1$): $S_n = a \times \frac{r^n - 1}{r - 1}$
- Infinite Geometric Series Sum ($|r| < 1$): $S_{\infty} = \frac{a}{1 - r}$
- Sum of first n natural numbers = $n \times \frac{n+1}{2}$
- Sum of squares of first n natural numbers = $\frac{n \times (n+1) \times (2n+1)}{6}$

Clocks & Calendars

- Angle between hands = $|30H - 5.5M|$
- $\frac{\text{Hands overlap exactly every 65 5}}{11 \text{ minutes}}$
- Leap year has 366 days (2 odd days), Ordinary year has 365 days (1 odd day)
- Odd days in 100 years = 5, 200 years = 3, 300 years = 1, 400 years = 0

Simple & Compound Interest

- Simple Interest (SI) = $\frac{P \times R \times T}{100}$
- Compound Interest Amount (A) = $P \times (1 + \frac{R}{100})^T$
- Compound Interest (CI) = A − P
- Difference between CI and SI for 2 Years = $P \times (\frac{R}{100})^2$
- If compounded half–yearly, Rate = $\frac{R}{2}$, Time = 2T

REASONING APTITUDE

Number & Letter Series

- Primary checking progression sequence: 1st Differences –> 2nd Differences –> Constant Ratio Multiplication
- Standard Series Models: n^2, n^3, $(n^2 + n)$, $(n^2 - 1)$, Fibonnaci additions
- A–Z letter conversions to numbers range from 1 to 26 (EJOTY shortcut: 5, 10, 15, 20, 25)

Coding-Decoding

- Linear Letter Shift Code Position: New Index = (Original Index + Shift Value) mod 26
- Mirror Alphabet Reverse Position mapping rule: Opposite Position = 27 − Current Position

Blood Relations

- Structural mapping shortcuts: $\frac{\text{Grandfather's only son}}{\text{daughter}} = \frac{\text{Father}}{\text{Mother}}$
- Parent's sibling = $\frac{\text{Uncle}}{\text{Aunt}}$ | Sibling's child = $\frac{\text{Nephew}}{\text{Niece}}$
- Pointer statements execution rule: Read from right to left starting at the final personal pronoun ('my')

Directions & Distances

- Turning Rules: Right Turn = 90 degrees Clockwise | Left Turn = 90 degrees Counter–Clockwise
- Shortest straight path displacement = Square Root of (Net Horizontal² + Net Vertical²)
- Shadow property: Morning shadow falls completely West | Evening shadow falls completely East

Seating Arrangements

- Linear Position conversion: Position from Right = Total Items − Position from Left + 1
- Items sandwiched between index positions X and Y = $|Y - X| - 1$
- Circular table orientation (facing center): Left movement = Clockwise | Right movement = Counter–Clockwise

Syllogism

- All A are B implies subset grouping layout: A is entirely contained inside B ($A \subseteq B$)
- Some A are B implies intersecting layout: Circle A overlapping Circle B
- No A is B implies completely disjoint layout: Isolated distinct circles
- Validity condition: A conclusion is true if and only if it holds across every possible Venn layout configuration

Clocks & Calendars

- Angle between Clock Hands = $|\, 30 \times H - 5.5 \times M \,|$ (If angle > 180, evaluate 360 − angle)
- Hands overlapping repetition period = $\frac{\text{Every 65 and 5}}{11 \text{ minutes}}$
- Odd days calculation system: Ordinary Year = 1 odd day | Leap Year = 2 odd days
- Century cycles odd days mapping: 100y = 5, 200y = 3, 300y = 1, 400y = 0

VERBAL APTITUDE

Reading Comprehension

- Main Idea checking boundary: Evaluate the structural introductory or summarizing lines of the paragraph
- Inference testing constraint: Hypotheses must map cleanly to explicit evidence inside the passage limits

Error Spotting & Sentence Correction

- Singular indefinite pronouns (Each, One, Either, Neither, Everyone) must bind with Singular Verbs
- 'One of the + [Plural Noun]' structural sequence requires a Singular Verb
- Prepositions mandate Objective Pronouns (e.g., Use 'between you and me' instead of 'you and I')
- $\frac{\text{Subjunctive}}{\text{Hypothetical expressions require conditional plural past tenses}}$

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Vocabulary

- $\frac{\text{Prefix}}{\text{Root translations}}$: bene– (good), mal– (bad), omni– (all), uni– (one), loqu– (speech)

Para Jumbles & Cloze Test

- Opening sentence constraints: Cannot begin with cross–referencing markers, linking pronouns, or conclusive transition words
- Logical Connector clues: Cause–Effect –> 'therefore', Contrast –> 'however', Addition –> 'moreover'

Para Jumbles

- Find the opening sentence: it introduces a topic without back–references (no 'this', 'however', 'therefore' as first word)
- Find the closing sentence: it concludes or summarises, often using 'thus', 'therefore', 'finally'
- Link pronouns to their antecedents: 'it' must follow the noun it refers to
- Connector words signal order: 'however'=contrast, 'moreover'=addition, 'therefore'=result, 'for example'=illustration
- Chronological or cause–effect chains: identify if the passage follows a timeline or argument structure

Reading Comprehension

- Main idea = what the entire passage is about, not just one paragraph
- Inference = what is implied but not directly stated — must be supported by the passage
- Tone clues: 'unfortunately', 'alarmingly' = $\frac{\text{concerned}}{\text{critical; 'remarkably'}}$, 'impressively' = $\frac{\text{positive}}{\text{appreciative}}$
- Vocabulary in context: replace the word with each option and check which fits the sentence meaning
- Elimination strategy: rule out options that are too extreme, out of scope, or contradict the passage

Cloze Test

- Read the full paragraph first before filling any blank
- The blank's word must fit grammatically $\frac{\text{noun/verb/adjective}}{\text{adverb}}$
- The blank's word must fit contextually $\frac{\text{positive/negative}}{\text{neutral tone}}$
- Collocations matter: 'make a decision' not 'do a decision'; 'strong coffee' not 'powerful coffee'
- Connectors reveal relationship: 'although'=contrast, 'since'= $\frac{\text{reason}}{\text{time}}$, 'despite'=contrast with noun

ADVANCED APTITUDE

Algebra

- Quadratic solution formula ($ax^2 + bx + c = 0$): $x = \frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$
- Sum of quadratic roots = $-\frac{b}{a}$ | Product of quadratic roots = $\frac{c}{a}$
- Two–digit integer expansion rule: Value of number with tens digit 't' and units digit 'u' = $10t + u$

Geometry & Mensuration

- 2D Area properties: Circle = πr^2, Triangle = $0.5 \times b \times h$, Square = $\frac{d^2}{2}$ (where d is the diagonal)
- 3D Volume metrics: Cylinder = $\pi r^2 h$, Cone = $\frac{1}{3}\pi r^2 h$, Sphere = $\frac{4}{3}\pi r^3$
- Pythagorean Theorem: Hypotenuse² = Base² + Perpendicular²

Data Sufficiency & Logical Puzzles

- Data Sufficiency standard checking sequence: Check (1) alone –> Check (2) alone –> Check Combined –> Conclude
- Sufficiency criteria: A statement is sufficient if and only if it produces exactly one unique outcome numerical answer

Data Interpretation

- Percentage Share = $\frac{\text{Value}}{\text{Total}} \times 100$
- Percentage Change = $\frac{\text{Final} - \text{Initial}}{\text{Initial}} \times 100$
- Average = $\frac{\text{Sum of Observations}}{\text{Number of Observations}}$
- Ratio A:B = $\frac{\text{Value of A}}{\text{Value of B}}$